

Joint Design of Transmit Waveform and Spatial Filter for MIMO Radar ISRJ Mitigation

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Abstract—This paper proposes a joint design of transmit waveform and spatial filter for multiple-input-multiple-output (MIMO) radar to mitigate interrupted sampling repeater jamming (ISRJ). Addressing the limitations of single-waveform phased arrays, we formulate a signal-to-interference-plus-noise ratio (SINR) maximization problem subject to the constant modulus constraint (CMC) to ensure hardware feasibility. To resolve the resulting non-convex optimization, we develop an efficient iterative algorithm integrating the majorization-minimization (MM) framework and the alternating direction method of multipliers (ADMM). The proposed method alternately optimizes the receive filter via the MVDR criterion and the transmit waveform via MM-ADMM, ensuring monotonic convergence. Simulation results demonstrate that the algorithm rapidly forms deep spatial nulls towards jammers. Compared to conventional linear frequency modulation (LFM) baselines, the proposed approach significantly improves output SINR and ISRJ suppression performance.

Index Terms—Alternating direction method of multipliers (ADMM), constant modulus constraint, interrupted sampling repeater jamming (ISRJ), majorization-minimization (MM), MIMO radar, non-convex optimization, waveform design.

I. INTRODUCTION

IN modern electronic warfare, interrupted sampling repeater jamming (ISRJ) poses severe threats to coherent radars by generating realistic false targets via digital radio frequency memory (DRFM) [1], [2]. Unlike phased arrays constrained by single-waveform transmission [3], multiple-input-multiple-output (MIMO) radar exploits orthogonal waveforms to expand the degrees of freedom (DoF) [4]. This spatial-waveform diversity enables joint transmitter-receiver optimization, offering a robust paradigm for ISRJ mitigation.

Minimum variance distortionless response (MVDR) is fundamental technique for suppressing spatial sidelobe jamming. A common approach involves maximizing the signal-to-

interference-plus-noise ratio (SINR) under various constraints, solved via methods such as alternating direction penalty method (ADPM) [5] and majorization-minimization (MM) [6]. Among these, the constant modulus constraint (CMC) is essential for ensuring high power-added efficiency [7] and robust detection. To address the resulting non-convex optimization, researchers employ cyclic algorithms [8], [9] and alternating direction method of multipliers (ADMM) [10].

In this paper, we propose a joint transmit waveform and spatial filter design to mitigate spatial-domain ISRJ for MIMO radar. We formulate the echo and jamming models to establish an SINR maximization problem, imposing a CMC to ensure hardware feasibility. The problem is solved by alternately optimizing the spatial filter and the transmit waveform. Specifically, the spatial filter is obtained by solving an unconstrained Rayleigh quotient problem. For the waveform optimization, the majorization-minimization (MM) framework is employed to handle the non-convex objective function, and the ADMM algorithm is used to decouple the problem into an unconstrained subproblem and a subproblem with CMC. Finally, the block coordinate descent (BCD) method solves the CMC subproblem.

Notations: In this paper, scalars are denoted by italic letters (e.g., x), vectors by bold lowercase letters (e.g., $\mathbf{x}$), and matrices by bold uppercase letters (e.g., $\mathbf{X}$). The superscripts $(\cdot)^T$, $(\cdot)^H$, and $(\cdot)^*$ represent the transpose, conjugate transpose, and complex conjugate, respectively. $\otimes$ denotes the Kronecker product, $\mathbb{E}[\cdot]$ is the expectation operator, while $\Re(\cdot)$ and $\text{Tr}(\cdot)$ stand for the real part and the trace of a matrix, respectively.

II. SYSTEM MODEL AND PROBLEM FORMULATION

Consider a ground-based radar system detecting an aircraft with a constant radial velocity v at height H , range R ,

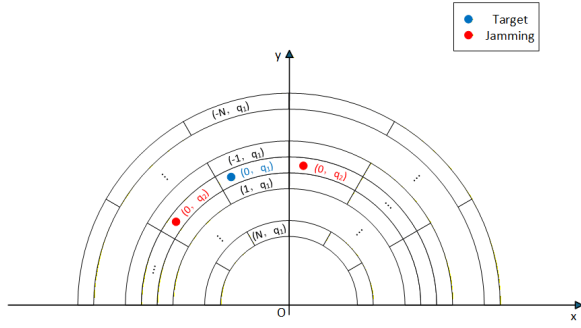


Fig. 1. Range-azimuth bins.

elevation Φ and azimuth Θ . To avoid the detection, the aircraft is equipped with an ISRJ jammer and several other aircraft with jammer and same velocity around it. By using the methods like [11], [12], the coordinate of the range-azimuth bin that the aircraft resist in range rings can be expressed as (k, q) (shown in Fig. 1), where k is the range relative to the target and q is the azimuth relative to the x-axis.

A. Target Model

For a moving target of interest located at azimuth θ_t , the transmit and receive steering vectors are defined as $\mathbf{a}_t(\theta_t) = [1, e^{j2\pi d_t \sin(\theta_t)/\lambda}, \dots, e^{j(N_t-1)2\pi d_t \sin(\theta_t)/\lambda}]^T$ and $\mathbf{a}_r(\theta_t) = [1, e^{j2\pi d_r \sin(\theta_t)/\lambda}, \dots, e^{j(N_r-1)2\pi d_r \sin(\theta_t)/\lambda}]^T$, where d_t and d_r denote the transmit antennas and receive antennas spacings respectively. The target echo matrix is expressed as

$$\mathbf{Y}_t = \alpha_t \mathbf{A}_t(\theta_t) \mathbf{S}, \quad (1)$$

where $\mathbf{A}_t(\theta_t) = \mathbf{a}_r(\theta_t) \mathbf{a}_t^T(\theta_t)$, while α_t denote the complex amplitude. Vectorizing (1), the echo signal becomes

$$\begin{aligned} \mathbf{y}_t &= \alpha_t (\mathbf{I}_N \otimes \mathbf{A}_t(\theta_t)) \mathbf{s} \\ &= \alpha_t \mathbf{v}_t \mathbf{s}, \end{aligned} \quad (2)$$

where $\mathbf{I}_N$ is the identity matrix of size N .

B. Jamming Model

Assume the target is escorted by N_j aircraft employing ISRJ. The signal intercepted by the i -th jammer is

$$\mathbf{j}_{r,i} = \mathbf{I}_N \otimes \mathbf{a}_t^T(\theta_{j,i}) \mathbf{s}, \quad (3)$$

which is retransmitted based on the ISRJ principle [13] as

$$\mathbf{j}_{t,i} = \mathbf{J}_{shift} (\mathbf{I}_N \otimes \mathbf{a}_t^T(\theta_{j,i})) \mathbf{s}, \quad (4)$$

where $\mathbf{J}_{shift}$ denotes the ISRJ transformation matrix.

For the i -th jammer located at $(\theta_{j,i}, p_i)$, the resulting echo is¹

$$\mathbf{y}_{j,i} = \alpha_{j,i} \mathbf{v}_{j,i} \mathbf{s}, \quad (5)$$

where $\alpha_{j,i}$ and $f_{j,i}$ are the amplitude and normalized Doppler frequency, respectively. The jamming steering vector is defined

¹For simplicity, we assume the jammers reside in the same range bin as the target.

as $\mathbf{v}_{j,i} = (\mathbf{f}_d(f_{j,i}) \otimes \mathbf{I}_N \otimes \mathbf{A}_t(\theta_{j,i})) \mathbf{j}_{shift,i}$. The transformation matrix $\mathbf{j}_{shift,i} = \text{Bdiag}(\mathbf{j}_{E,i}, \mathbf{j}_{E,i}, \dots) \in \mathbb{C}^{N_{N_t} \times N_{N_t}}$ consists of block diagonal elements

$$\mathbf{j}_{E,i} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{I}_K & \vdots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{I}_K & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0}_{q_i \times K} & \mathbf{0}_{q_i \times K} & \cdots & \mathbf{0}_{q_i \times K} \end{bmatrix}, \quad (6)$$

where $K = N_{j,s,i} N_t$ and the shift index is $q_i = (T_{j,s,i} - (N_{j,r,i} + 1)N_{j,s,i})N_t$.

Under this configuration, the composite received signal $\mathbf{y}$, comprising the target echo, jamming, and background noise, is expressed as

$$\mathbf{y} = \mathbf{y}_t + \sum_{i=1}^{N_j} \mathbf{y}_{j,i} + \mathbf{n}, \quad (7)$$

where $\mathbf{n}$ denotes the additive white Gaussian noise vector.

C. Objective Function Formulation

Consider a spatial filter $\mathbf{w} \in \mathbb{C}^{N_{N_r} \times 1}$ producing the output $\mathbf{r} = \mathbf{w}^H \mathbf{y}$. To suppress sidelobe jamming, we maximize the output SINR:

$$\text{SINR} = \frac{|\mathbf{w}^H \mathbf{y}_t|^2}{\mathbf{w}^H \mathbf{R}_I \mathbf{w}}, \quad (8)$$

where $\mathbf{R}_I = \mathbb{E}[\mathbf{y}_{un} \mathbf{y}_{un}^H]$ is the interference-plus-noise covariance matrix. Substituting the jamming model, $\mathbf{R}_I$ is expressed as

$$\mathbf{R}_I = \sum_{i=1}^{N_j} \alpha_{j,i}^2 \mathbf{v}_{j,i} \mathbf{s} \mathbf{s}^H \mathbf{v}_{j,i}^H + \alpha_n \mathbf{I}_{N_{N_r}}, \quad (9)$$

where α_n denotes the noise power and $\mathbf{y}_{un}$ represents the interference and noise components.

D. Optimization Problem Under CMC

The joint design problem is formulated to maximize the SINR for sidelobe jamming suppression. To guarantee hardware feasibility, we impose a CMC on the transmit waveform. The resulting optimization problem is expressed as:

$$\begin{aligned} \max_{\mathbf{w}, \mathbf{s}} \quad & \text{SINR}(\mathbf{w}, \mathbf{s}) \\ \text{s.t.} \quad & |\mathbf{s}(n)| = 1, \quad \forall n. \end{aligned} \quad (10)$$

The optimization problem in (10) is non-convex due to the fractional objective function coupled with the strict equality constraint of the CMC. Consequently, a direct closed-form solution is intractable. To address this, we propose an efficient iterative algorithm in the following section, which decouples the design of the receive filter and the transmit waveform.

III. MM-ADMM BASED ALGORITHM FRAMEWORK

Directly solving (10) is intractable due to the coupling between $\mathbf{s}$ and $\mathbf{w}$. Therefore, we decompose the optimization into two sub-problems and solve them iteratively.

A. Optimization of Receive Filter $\mathbf{w}$

Unlike classical MIMO-STAP, the interference covariance matrix $\mathbf{R}_I$ here depends on $\mathbf{s}$, creating a coupling effect. Following [14], we reduce the complexity by optimizing $\mathbf{w}$ with a fixed $\mathbf{s}$, ignoring the coupled Integrated Sidelobe Level Constraint (ISLSC) without compromising efficacy. The sub-problem becomes:

$$\max_{\mathbf{w}} \frac{|\mathbf{w}^H \mathbf{y}_t|^2}{\mathbf{w}^H \mathbf{R}_I \mathbf{w}}. \quad (11)$$

This is a standard Rayleigh quotient problem, equivalent to the Minimum Variance Distortionless Response (MVDR) formulation:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \mathbf{w}^H \mathbf{R}_I \mathbf{w} \\ \text{s.t.} \quad & \mathbf{w}^H \mathbf{y}_t = 1, \end{aligned} \quad (12)$$

which yields the analytical solution:

$$\mathbf{w}^{opt} = \frac{\mathbf{R}_I^{-1} \mathbf{v}_t \mathbf{s}}{\mathbf{s}^H \mathbf{v}_t^H \mathbf{R}_I^{-1} \mathbf{v}_t \mathbf{s}}. \quad (13)$$

B. MM-Based Waveform Optimization

Substituting (13) into (10), the optimization with respect to $\mathbf{s}$ becomes:

$$\begin{aligned} \min_{\mathbf{s}} \quad & -\mathbf{s}^H \mathbf{v}_t^H \mathbf{R}_I^{-1} \mathbf{v}_t \mathbf{s} \\ \text{s.t.} \quad & |\mathbf{s}(n)| = 1. \end{aligned} \quad (14)$$

Since $\mathbf{R}_I$ depends on $\mathbf{s}$, the problem is non-convex. We apply the MM framework to linearize the objective.

Lemma 1 ([15]): Let $f(\mathbf{s}, \mathcal{S})$ denote the objective of (14) with $\mathcal{S} = \mathbf{s} \mathbf{s}^H$. A majorizer of $f(\mathbf{s}, \mathcal{S})$ at the l -th iteration is constructed as:

$$\begin{aligned} g_1(\mathbf{s}; \mathbf{s}_{(l)}) = & -2\Re(\mathbf{z}_{(l)}^H \mathbf{s}) + 2\text{Tr}(\mathbf{P}_{(l)} \mathcal{S}) \\ & - 2\text{Tr}(\mathbf{P}_{(l)} \mathcal{S}_{(l)}) - f(\mathbf{s}_{(l)}, \mathcal{S}_{(l)}), \end{aligned} \quad (15)$$

where the auxiliary variables are defined as:

$$\begin{aligned} \mathbf{z}_{(l)} &= \mathbf{v}_t^H \mathbf{R}_{I,(l)}^{-1} \mathbf{v}_t \mathbf{s}_{(l)}, \\ \mathbf{P}_{(l)} &= \sum_{i=1}^{N_j} \alpha_{j,i} \mathbf{Q}_{(l),i}^H \mathcal{S}_{(l)} \mathbf{Q}_{(l),i}, \\ \mathbf{Q}_{(l),i} &= \mathbf{v}_t^H \mathbf{R}_{I,(l)}^{-1} \mathbf{v}_{j,i}. \end{aligned} \quad (16)$$

Applying Lemma 1 and discarding constant terms, the majorized objective reduces to:

$$\min_{\mathbf{s}} \quad \mathbf{s}^H \mathbf{P}_{(l)} \mathbf{s} - \Re(\mathbf{z}_{(l)}^H \mathbf{s}). \quad (17)$$

Consequently, the optimization problem at the l -th iteration is formulated as:

$$\begin{aligned} \min_{\mathbf{s}} \quad & \mathbf{s}^H \mathbf{P}_{(l)} \mathbf{s} - \Re(\mathbf{z}_{(l)}^H \mathbf{s}) \\ \text{s.t.} \quad & |\mathbf{s}(n)| = 1. \end{aligned} \quad (18)$$

The proposed MM procedure is summarized in Algorithm 1.

Algorithm 1 MM-based Algorithm for Solving (14)

- 1: **Input:** Initial waveform $\mathbf{s}_{(0)}$, max iterations l_{max} .
 - 2: **Output:** Optimized waveform $\mathbf{s}^*$.
 - 3: Initialize $l = 0$.
 - 4: **while** not converged **and** $l < l_{max}$ **do**
 - 5: Update auxiliary parameters $\mathbf{z}_{(l)}$ and $\mathbf{P}_{(l)}$ via (16).
 - 6: Solve (18) to obtain $\mathbf{s}_{(l+1)}$.
 - 7: Update $l = l + 1$.
 - 8: **end while**
-

C. ADMM Implementation

To solve (18), we employ the ADMM framework by introducing an auxiliary variable $\mathbf{x}$. The reformulated problem is [16]–[18]:

$$\begin{aligned} \min_{\mathbf{s}, \mathbf{x}} \quad & \mathbf{s}^H \mathbf{P}_{(l)} \mathbf{s} - \Re(\mathbf{z}_{(l)}^H \mathbf{s}) \\ \text{s.t.} \quad & \mathbf{s} = \mathbf{x}, \quad |\mathbf{x}(n)| = 1. \end{aligned} \quad (19)$$

The corresponding augmented Lagrangian function is given by:

$$\begin{aligned} \mathcal{L}(\mathbf{s}, \mathbf{x}, \boldsymbol{\lambda}) = & \mathbf{s}^H \mathbf{P}_{(l)} \mathbf{s} - \Re(\mathbf{z}_{(l)}^H \mathbf{s}) \\ & + \Re(\boldsymbol{\lambda}^H (\mathbf{x} - \mathbf{s})) + \frac{\rho}{2} \|\mathbf{x} - \mathbf{s}\|^2, \end{aligned} \quad (20)$$

where $\boldsymbol{\lambda}$ is the dual vector and ρ denotes the penalty parameter. Intuitively, introducing the auxiliary variable $\mathbf{x}$ decouples the intractable optimization into two manageable sub-problems: $\mathbf{s}$ handles the unconstrained objective minimization, while $\mathbf{x}$ strictly enforces the non-convex constant modulus constraint. The update steps operate as an alternating consensus process. Driven by the dual variable $\boldsymbol{\lambda}$'s penalty mechanism, $\mathbf{x}$ projects the signal to satisfy hardware limits, and $\mathbf{s}$ optimizes interference suppression. They iteratively converge to a consensus, ensuring both high performance and hardware feasibility.

1) *Update of Auxiliary Variable $\mathbf{x}$:* In the k -th iteration, the optimization for $\mathbf{x}$ is separable due to the element-wise CMC. Via BCD, it decomposes into N parallel sub-problems:

$$\begin{aligned} \min_{\mathbf{x}(n)} \quad & \Re(\boldsymbol{\lambda}_{(k-1)}(n)(\mathbf{x}(n) - \mathbf{s}_{(k-1)}(n))) \\ & + \frac{\rho_{(k-1)}}{2} |\mathbf{x}(n) - \mathbf{s}_{(k-1)}(n)|^2 \\ \text{s.t.} \quad & |\mathbf{x}(n)| = 1 \end{aligned} \quad (21)$$

which admits the closed-form analytical solution:

$$\mathbf{x}_{(k)}(n) = -\frac{\boldsymbol{\lambda}_{(k-1)}(n) - \rho_{(k-1)} \mathbf{s}_{(k-1)}(n)}{|\boldsymbol{\lambda}_{(k-1)}(n) - \rho_{(k-1)} \mathbf{s}_{(k-1)}(n)|}. \quad (22)$$

2) *Update of Primal Variable $\mathbf{s}$:* The sub-problem for $\mathbf{s}$ is an unconstrained quadratic minimization:

$$\min_{\mathbf{s}} \quad \mathcal{L}(\mathbf{s}, \mathbf{x}_{(k)}, \boldsymbol{\lambda}_{(k-1)}, \rho_{(k-1)}). \quad (23)$$

Since (23) is convex, setting the gradient to zero yields the update rule:

$$\mathbf{s}_{(k)} = (2\mathbf{P}_{(l)} + \rho_{(k-1)} \mathbf{I}_N)^{-1} (\mathbf{z}_{(l)} + \boldsymbol{\lambda}_{(k-1)} + \rho \mathbf{x}_{(k)}). \quad (24)$$

Algorithm 2 ADMM-based Algorithm for Solving (18)

- 1: **Input:** Initial $\mathbf{s}_{(0)}$, parameters k_{max}, λ_{max} .
 - 2: **Output:** Optimized $\mathbf{s}^*$.
 - 3: Initialize $k = 0$.
 - 4: **while** $\|\mathbf{x}_{(k)} - \mathbf{s}_{(k)}\|_2 > \epsilon$ **and** $k < k_{max}$ **do**
 - 5: Update $\mathbf{x}_{(k+1)}$ via (22).
 - 6: Update $\mathbf{s}_{(k+1)}$ via (24).
 - 7: Update $\boldsymbol{\lambda}_{(k+1)}$ via (25).
 - 8: $k = k + 1$.
 - 9: **end while**
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TABLE I
SYSTEM PARAMETERS

Parameter	Value
Tx/Rx Antennas (N_t, N_r)	3, 4
Antenna Spacing (d_t, d_r)	2
Code Length (N)	32
Pulse Number (M)	4
PRF / Carrier Freq. (f_c)	2000 Hz / 10^8 Hz
Number of Jammers	3
Target Azimuth (θ_t) / Amp. (α_t)	$\pi/12$, 1
Step Size (ρ)	100
Penalty Upper Bound (λ_{max})	10000

TABLE II
ISRJ PARAMETERS

Parameter	J1	J2	J3
Codes per segment (N_{js})	1	2	2
Repetitions (N_{jr})	2	2	3
Sampling Interval (T_{js})	4	8	4
Azimuth (θ_j)	$-\frac{5\pi}{18}$	$-\frac{\pi}{18}$	$\frac{2\pi}{9}$
Amplitude (α_j)	1	1	1

3) *Update of Dual Variable $\boldsymbol{\lambda}$:* To ensure numerical stability, the dual variable is updated with a projection bound:

$$\boldsymbol{\lambda}_{(k)} = \begin{cases} \hat{\boldsymbol{\lambda}}_{(k)}, & \text{if } \max(\hat{\boldsymbol{\lambda}}_{(k)}) < \lambda_{max} \\ \hat{\boldsymbol{\lambda}}_{(k)} \frac{\lambda_{max}}{\max(\hat{\boldsymbol{\lambda}}_{(k)})}, & \text{otherwise,} \end{cases} \quad (25)$$

where the intermediate update is $\hat{\boldsymbol{\lambda}}_{(k)} = \boldsymbol{\lambda}_{(k-1)} + \rho_{(k)}(\mathbf{x}_{(k)} - \mathbf{s}_{(k)})$, and λ_{max} denotes the preset upper bound.

4) *Convergence Criteria:* The algorithm terminates when the primal residual $\|\mathbf{x}_{(k)} - \mathbf{s}_{(k)}\|_2$ falls below a predefined threshold ϵ or the maximum iteration count k_{max} is reached. The ADMM procedure is summarized in Algorithm 2.

IV. NUMERICAL RESULTS

This section demonstrates the algorithm's feasibility via numerical simulations. The transmit waveforms are initialized with random sequences. System and ISRJ parameters are detailed in Table I and Table II, respectively².

Fig. 2 illustrates the convergence behavior over 50 outer loops with varying step sizes ρ . The algorithm exhibits robust convergence for $\rho \in \{1, 10, 100\}$, reaching a stable SINR of 36.52 dB within just two iterations. Average runtimes for these values are 0.94 s, 0.25 s, and 0.22 s, respectively. Conversely, larger ρ (e.g., 1000) significantly slows convergence (10 iterations, 0.17 s), while extremely large values (5000, 10000)

² T_{js} is measured in code elements.

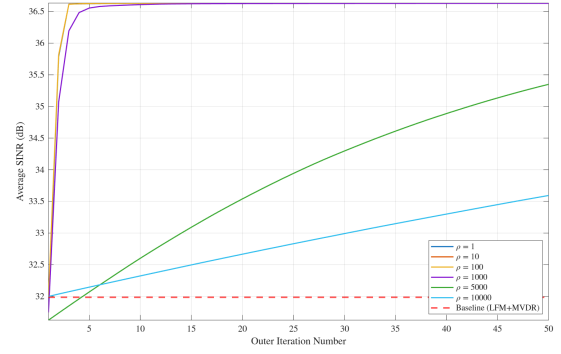


Fig. 2. Convergence curve under different penalty parameters ρ .

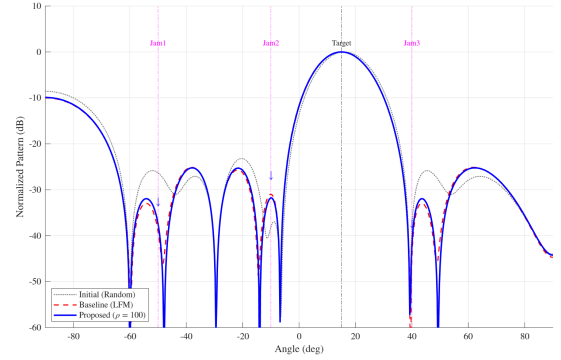


Fig. 3. Spatial response patterns: Proposed vs. LFM and Initial baselines.

fail to converge within the observed window. This indicates that while overly small ρ may degrade inner-loop ADMM efficiency, an appropriately selected ρ ensures rapid outer-loop convergence.

Fig. 3 depicts the spatial response pattern³. Deep nulls (below -30 dB) are adaptively formed in the directions of all three jammers. Compared to the initial random pattern, the proposed algorithm boosts SINR by approximately 4.6 dB. Notably, while the spatial response resembles the linear frequency modulation (LFM) baseline, the SINR gain is primarily achieved through the joint optimization of the transmit waveform, validating the efficacy of the proposed space-waveform design.

V. CONCLUSION

This paper proposes a joint space-waveform design framework for MIMO radar against ISRJ. The core innovation lies in exploiting MIMO diversity to transform the coupled SINR maximization under strict CMC into an efficient MM-ADMM paradigm, breaking the limitations of single-waveform phased arrays. In engineering practice, the strict CMC directly guarantees high power amplifier efficiency, while the rapid convergence and adaptive deep-nulling capability provide a robust and hardware-feasible anti-jamming solution against DRFM false targets.

³Baseline: LFM waveform with MVDR; Initial: Random sequence with optimal $\mathbf{w}$ from (13).

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